J Happiness Stud
DOI 10.1007/s10902-017-9882-x

RESEARCH PAPER

Happiness at Work Scale: Construction and Psychometric Validation of a Measure Using Mixed Method Approach

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Abstract Using a mixed method approach we conducted three studies to construct a multidimensional measure of happiness at work. In Study 1, a qualitative content analysis using Atlas.ti 7 offered support for the proposed, a priori deductive model which also discriminated between the construct of happiness at work and happiness in personal life. Further, a principal axis factoring of the responses consisting of 539 working people (Study 2) yielded four factors reflecting the equal role of organizational and human aspects of the workplace happiness, i.e. supportive work experiences, unsupportive work experiences, flow and intrinsic motivation and work repulsive feelings. In Study 3, a Confirmatory Factor Analysis testing the reflective versus formative structure of the proposed scale supported a four-factor reflective model over a series of 13 competing models. Moreover, the scale showed statistically significant convergent and discriminant validity. Possible applications of the scale in predicting happiness styles and enhancing the experience of happiness at work has been presented at the end.

Keywords Happiness at work scale · Happiness styles · Subjective well-being

1 Introduction

There seems to be a steep rise of popular as well as research interests in the construct of happiness at work (Fisher 2010; Pryce-Jones 2010) as it brings a range of positive benefits for individuals as well as for organizations. Happiness at work (HAW) has been found to be associated with greater career success, earning, job performance and helping others at

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Published online: 27 April 2017

work (Boehm and Lyubomirsky 2008). Happy employees are twice as productive, six times more energized, take the only 1/10th of sick-leaves, and intends to stay twice as long in the organizations as compared to an unhappy employee (Pryce-Jones and Lindsay 2014).

1.1 Conceptual Operationalization of the Construct

In the research literature, the idea of HAW has been drawn on the happiness research mainly in the field of psychology (Diener et al. 1993) and economics (Krause 2014). Happiness is often considered synonymous with subjective well-being and defined as a state of the high level of satisfaction with life coupled with high level of positive affect and low level of negative affect (Carr 2004; Sheldon and Lyubomirsky 2004). However, the construct of subjective well-being, coined by Diener, denotes only an aspect of happiness that can be empirically measured. Hence, some researchers consider subjective well-being as a scientific synonym of happiness (Pawelski and Gupta 2011). However, happiness is part of everyday human experience and discourse. Thus, it will necessarily have fuzzy boundaries and varying definitions (Diener 1984). For example, researchers have stressed upon good luck and favourable external conditions (Oishi et al. 2013), coping resources (Cohn et al. 2009), time and other orientation (Aaker et al. 2010) while understanding and defining happiness. There is no unanimity over the content, definition and measures of happiness (Agbo et al. 2012) with some researchers considering indicators of subjective well-being, affect, quality of life denoting happiness (Easterlin 2004), while others highlighting the need to differentiate between them (Bruni and Porta 2005).

Therefore, while conceptualizing HAW we have kept in mind the important, if not universal, measurement aspect of subjective well-being along with keeping our understanding of HAW simple, experiential and grounded in data (study 1). We conceptualize HAW as an experience of subjective well-being at work that involves interaction between individual employee experiences as well as organizational experiences. We operationally define HAW as an experience of subjective well-being at work reflected through a high amount of positive individual (e.g., highly valuing one's work, feeling engaged to work) and organizational (e.g., providing supportive work environment) experiences and low amount of negative individual and organizational experiences. Based on this we propose a priori framework (see Fig. 1) suggesting that a high amount of HAW will bring a high amount of happiness-reflecting employee (e.g., feeling highly engaged to one's work) and organizational experiences (e.g., satisfaction with professional support and training received). Similarly, a high amount of happiness means low amount of happiness reducing (thus, unhappiness increasing) employee-level (e.g., strongly disliking one's work) and organizational experiences (e.g., the absence of avenues for meaningful work that impact others).

2 Review of Literature

Fisher (2010) has offered three levels of analysis of HAW, i.e., personal, collective and transient/state laying emphasis on HAW as an individual experience affected by the work group dynamics. She also takes into account the momentary, but profound, experiences that play a vital role in this entire relationship within the broader organizational context. The important personal level variables, according to her, related to HAW are job satisfaction, dispositional affect, affective organizational commitment, job involvement, work

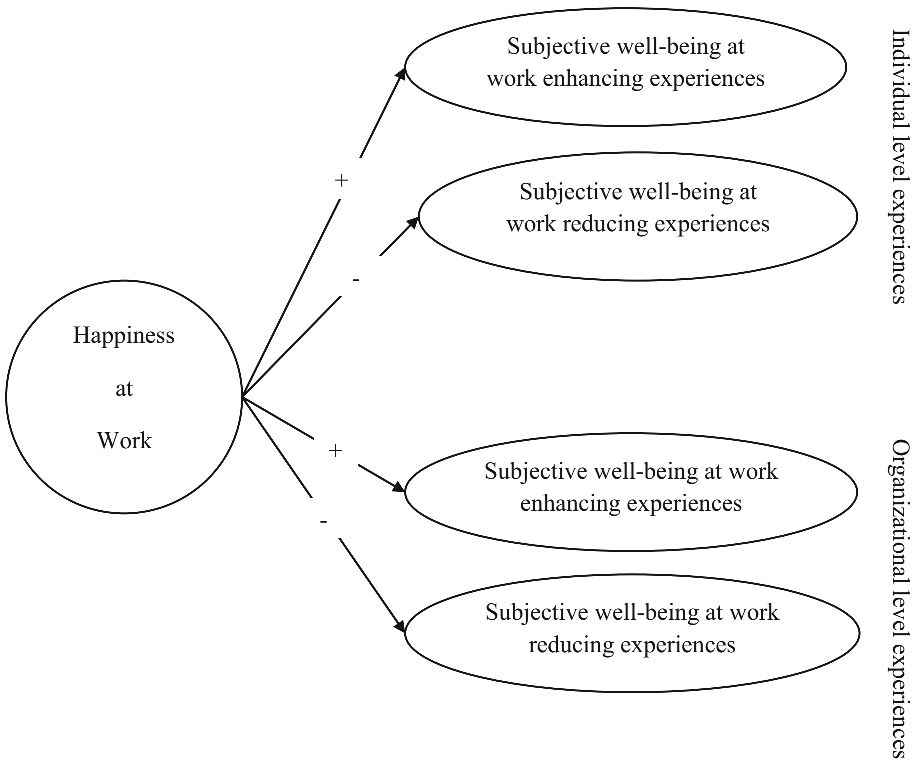

Fig. 1 A priori operationalization and factor structure of the proposed scale. A high amount of happiness at work will be reflected in high amount of those individuals and organizational experiences that increased subjective well-being of employees at work

engagement, the typical mood at work, thriving, vigour and flourishing, and affective well-being at work. Many of these individual level constructs extrapolate at collective and transient stage affecting the happiness at that particular level.

Fisher's framework seems appealing and comprehensive but gives an overwhelming emphasis on job satisfaction and affect variables at various levels. For example, affect appears as a dispositional variable, mood state, affective well-being as well as a sort of affective commitment towards organizations. Fisher has traced the construct of HAW to its positive psychological roots, and, thus this outcome doesn't seem surprising. However, the important point is that the study gave us a background, helped us to begin with a broad deductive framework while inductively accommodating new constructs from other studies by saturating the existing constructs. As the aim of the current research was to measure HAW at individual level only so we ignored the group and transient state level constructs. Finally, based on Fisher we selected dimension of job satisfaction, work engagement, thriving and vigour, flow, and amount of positive affect for item generation (see Table 1).

Furthermore, to inductively derive factors from employee's perspective we carried out a qualitative study (Study 1) in which 260 employees offered their feedback about what makes them happy at work in response to an online survey. Apart from reflecting the dimensions outlined earlier we additionally got dimensions of supportive work

Table 1 Dimensions of the proposed scale

s. no.	Dimensions	Operational definition (extent to which an employee)	Sample item	Source
1	Job satisfaction	…scores high on contentment with job	"I don't get sufficient credit for my achievements.*"	Fisher (2010)
2	Work engagement	…enjoys work and remains devoted to it	"I feel deeply involved in my work."	Fisher (2010)
3	Thriving and vigour	…feels flourishing and energetic at work	"My work provides me sufficient opportunity to realize my full potential."	Fisher (2010)
4	Flow	…feels absorbed in his work	"When I start doing my work I forget everything else."	Fisher (2010)
5	Affect	… feels higher amount of positive emotions and lowers amount of negative emotions	"My work makes me smile."	Fisher (2010)
6	Supportive work experience	…respects top leaders and managers for creating supportive work environment	"My organization provides all necessary training and information to complete work on time."	Expert feedback, qualitative study
7	Team experiences	…perceives inclusive team culture in organization	"My organization does not have proper guidelines to regulate team behavior and the work that requires collective effort.*"	Expert feedback, qualitative study
8	Workplace hygiene	…perceives workplace safe, uncluttered and hygienic	"Many of my colleagues have improper sanitary habits.*"	Qualitative study
9	Job security	…feels secure at his/her job	"I feel secure at my job."	Expert feedback, qualitative study
10	Impact on others	…perceives work has a significant impact on the lives of others	"My work has a positive impact on the lives of my company customers and other people."	Hackman and Oldham (1976), expert feedback

* Denote the negatively worded items of the scale

experiences, team experiences, feeling of workplace hygiene, perceived job security and the impact of work on lives of others. These dimensions are briefly discussed below.

Job satisfaction is defined as an individual's positive attitude toward job (Robbins and Judge 2012). Happy individuals are reported to be "more loving, forgiving, trusting, energetic, decisive, creative, sociable, and helpful" (cited from Myers 2000, p. 58) which may lead to employees develop a more favourable attitude toward their work. A high level of energy, creativity and enthusiasm can motivate the employees to devote themselves more to work (given that organization provides sufficient avenues for the same). Furthermore, happy employees show more sensitivity to opportunities at work and remain confident and optimistic about their work (Bakker and Demerouti 2008). This would increase the *work engagement* of the happy employees and sometimes an employee may feel fully devoted or absorbed in work. As compared to this an unhappy employee is less likely to feel engaged or to flourish at work and more likely to feel stressed and exhausted leading to a diminished sense of *thriving and vigour* at work. This will also reduce the job performance of the employee (Carmeli et al. 2009).

Related to this is the concept of *flow*, defined as a state of experiencing peak enjoyment, energetic focus, and creative concentration when an employee is completely immersed in the job. Based on the literature cited earlier we believe that a happy employee will experience more creativity, energy, focus and enjoyment at work leading to the experience of flow. However, flow has been considered the main source or ingredient of happiness (Csikszentmihalyi 2013) which is likely to reflect a formative influence of flow on happiness. Nonetheless, happiness in the context of flow research has been variedly, sometimes very differently, defined as compared to the main idea of scientific happiness. For example, Bakker (2008) in his measurement model of flow considered work enjoyment as a dimension of flow which is close to happiness but not exactly happiness as it is understood in positive psychology since the component of satisfaction, although seems implied, but objectively missing in his flow inventory. This indicates that researchers believe that flow brings experiences similar to happiness (e.g., work enjoyment) but may not lead to exactly what we call scientific happiness. A high energy, enthusiasm and heightened focus brought by happiness (Bakker and Demerouti 2008) may itself lead to the experience of flow. To clarify the reflective versus formative relation between flow and happiness, we have tested both reflective and formative measurement model and found support in favour of the reflective influence of HAW on flow (see Table 4). Finally, the dimensions of work engagement, thriving and vigour and flow may appear overlapping, but Spreitzer and Porath (2014) has drawn fine distinctions among them.

Affect at work is another important variable in this context as happiness is often experienced in the form of positive affective states. A high-level of positive affect coupled with high level of satisfaction has been an important component of happiness in positive psychology literature (Carr 2004). We believe that a high amount of HAW will be reflected through a high amount of positive affect and low amount of negative affect. Furthermore, employees spend a significant amount of time in organizations interacting with co-workers, working in teams and meeting the expectations set out by their leaders. Employees develop different subjective evaluations of organizational support and team experiences depending upon their feelings and the level of satisfaction at work. As happy people are more optimistic and forgiving, we hypothesize that happy employees will value their organizational and team experiences more favourably while an unhappy employee will essentially have complains and negativity for the same. Hence, we took supportive work experiences and *team experiences* as two other aspects of HAW.

Workplace hygiene is a major issue at the workplace and according to a survey around 57% people at work judge a coworker based on the extent to which the coworker keeps the workplace clean or dirty (Goudreau 2012). Unhappy people show less keenness to plan and are less organized (White 2014) which could be reflected in the form of the messy desk or unhygienic habits. Happy people are better workers (McKee 2014) and this will have implication for workplace hygiene too. We do not deny the possibility of a formative role of workplace hygiene on HAW, but we focus on the important relationship between two variables and the implication that HAW will have for workplace hygiene. Furthermore, since happy people are more socially outgoing, confident and sensitive to new opportunities they are more likely to feel secure (*job security*) and growing professionally. However, it's more customary to think that it is job security promising a regular stream of income that will lead to happiness but such influence of a regular job/income on happiness has often been less than satisfactory or mostly illusory (Kahneman et al. 2006).

The last dimension taken in the study was the *impact of work on others* which we found conceptually closer to the concept of task significance given in the Job Characteristics Model (Hackman and Oldham 1976). As pointed by Dunn et al. (2011) happiness is related to focusing on the experiences rather than material good, becoming more other-oriented, giving importance to the happiness of others and contributing to the lives of others. A genuine happiness will nudge an employee to look for work avenues that will contribute positively to the lives of others. The theory of authentic happiness (Seligman 2002) considers hedonic, desirable as well as meaningful pursuits important for happiness. PERMA model of well-being (Seligman 2011) emphasizes on positive emotion, engagement, relationship, meaning and accomplishment (PERMA). We find these ten dimensions addressing authentic happiness and PERMA model at work to a good extent.

3 Scale Development

3.1 Critique of the Existing Measures of HAW

The scale development process started with the search for existing scales on the HAW while avoiding the subjective well-being measures assessing happiness in general or personal life. We found that there has been focused attempt to develop a scales for measuring HAW mainly at professionals level but less so at the academic level. Happiness at Work Survey (Happiness Works Ltd. 2016) and SMILES Framework (Andrew 2011) fall into the category of scales developed by professionals. Noticeable scales with the academic approach are the Work Well-being Scale (Tatiane and Tamayo 2008), the Job-Related Affective Well-Being Scale (Van Katwyk et al. 2000) and iPPQ based HAW Scale (Lutterbie and Pryce-Jones 2013). There has been some amount of overlap between the core dimensions of HAW captured by these scales indicating increasing consensus on the content validity of the construct. However, the scales falling within professional category suffer from the lack of research rigour. Some of these scales (e.g., iPPQ) appears more of a popular measure and cannot be considered scientifically developed and validated scale of HAW.

3.2 Development of Dimensions and Item Writing

The ten dimensions of the proposed scale were identified based on the conceptual triangulation between literature review, expert feedback and a qualitative content analysis (Study 1). Since literature review and the feedback of two experts gave us sufficient conceptual framework to proceed with identifying key dimensions, therefore, we coded the qualitative data obtained in Study 1 using a combination of inductive and deductive approaches (Friese 2011, 2014) in Atlas.ti 7 software. First five out of total ten dimensions were based mainly on the literature review while the remaining five dimensions were based primarily on the expert feedback and the qualitative analysis (Study 1). Based on this we operationally defined all the ten dimensions and the details of the same are given in Table 1 along with sample item and relevant sources.

3.3 Proposed Factor Structure and Content Validity of the Scale

Following a priori framework (Fig. 1) we wrote both positive and negative items on each of the dimensions of HAW identified through review of the literature and qualitative analysis (study 1). We expected a hierarchical four factor model of HAW to measure the proposed construct. Later, the results of the EFA (Study 2) confirm the emergence of four factors which were validated based on the confirmatory analysis (Study 3).

3.4 Reliability, Convergent, and Discriminant Validity

To assess the convergent validity during the CFA we included only those items in a dimension which showed standardized factor loading greater than .7, with composite reliabilities of the dimensions greater than .7, and Average Variance Extracted (AVE) value more than .5 (Hair et al. 2013). Maximal reliability or Coefficient *H* was calculated to assess the stability of the dimension over repeated administrations. A construct with *H* coefficient more than .8 indicates good stability (Hancock and Mueller 2001). Coefficient *H* is considered a better measure of the construct reliability as compared to Cronbach's alpha coefficient as the later reflects the reliability of a composite only rather than the reliability obtained through measured indicators of the factor (Hancock and Mueller 2010). We also checked the correlations between the proposed scale scores and other standardized constructs like subjective well-being, positive emotion and negative emotion, satisfaction with life and optimistic life orientation (for psychometric details of these measures see study 3; for results of correlation analysis see Table 3). For discriminant validity, we checked if Maximum Shared Variance (MSV) values were less than AVE, along with whether square root of AVE values were greater than off-diagonal correlations (Fornell and Larcker 1981; Hair et al. 2013). The calculation for validity was done using the master validity plug-in AMOS (Gaskin and Lim 2016) and results are presented in Table 5.

3.4.1 Discriminant Validation of Reflective versus Formative Structure of the Proposed Scale

We also attempted to distinguish whether the reflective or formative structure of the proposed scale better fits to data using three well-studied constructs associated with happiness, i.e. affect (positive versus negative), satisfaction with life, and life orientation. Psychological constructs like happiness, personality, attitude are essentially reflective constructs in nature (Diamantopoulos and Siguaw 2006; Diamantopoulos and Winklhofer 2001). Thus, we hypothesized that relating a reflective model of HAW with reflective models of affect (positive versus negative), satisfaction with life, and life orientation will fit better to data as compared to relating a formative model of HAW with the same. For this, we took reflective measurements of affect, satisfaction with life, and life orientation and tested various combinations of measurement models by varying the proposed scale's measurement model between reflective and formative models (see Table 4 for results).

4 Study 1: Identifying Dimensions of HAW Scale Through Qualitative Inquiry

4.1 Method

An open-ended qualitative survey was used to collect information regarding events occurred within past one year that respondents associate with workplace happiness. Participants were also asked to mention the anticipated happiness related events at their workplace in the coming year. To differentiate between events that people relate with happiness at work in contrast to happiness in personal life the respondents were also asked to mention similar events in their past and future one years of life. The obtained data was qualitatively analyzed for the major themes underlying HAW. Based on the review of literature some provisional codes (Saldana 2015) were developed which were further refined and updated based on the codes grounded in qualitative data. A combination of deductive and inductive approach was used by the authors to code data as per the Noticing-Collecting-Thinking approach (Friese 2011, 2014). The coding process was initiated with a deductive framework proposed in the beginning (Fig. 1). The codes which were not found susceptible to provisional coding were developed inductively through open coding approach.

4.2 Sample

The qualitative survey was sent to 680 working people with the help of psychology undergraduate students of the University of Delhi. We received 260 usable responses making effective response rate of 38.23%. The sample consisted of 63% male and 37% female executive level employees from various parts of India.

4.3 Results and Discussion

The qualitative coding process led to the generation of total 118 open codes. The network view showed the emergence of different themes for HAW under the two broad deductive category of individual and organizational factors. However, it was noticed that while talking about HAW respondents also referred to unhappiness even when they were not explicitly asked about it. Some respondents related their unhappiness with a high workload, feeling job insecurity, transfer or relocation. Unhappy employees also made reference to office politics, bad work environment, disproportionate workload, an insufficient number of holidays, unfair treatment, stress, job uncertainties, long commuting distance, poor location, untidy and cluttered workplace, and unwarranted transfers.

The individual and organizational factors of happiness were saturated with inductively derived codes which were subsequently converted into categories. The one prominent organizational aspect signified toward a positive work experience which contained appreciation for organizational support/facilities, colleagues, and bosses. The other aspects were networking and growth of opportunities, salary, promotion, paid leaves, holidays, tours and vacations, job security and perception of top leaders including leadership ethics, perceived fairness toward employees and the fair decision-making process. Participants also mentioned individual level intrinsic experiences such as personal achievements that boosted the self-esteem of the employees, their engagement with work and the extent to which their work impacted the lives of others including company customers, colleagues, their family members as well as society in general. Matching with the *zeitgeist* employees also cited mastering new skills and learning new things related to HAW.

In contrast to this, the most prominent themes related to happiness in life were the occurrence of social events like celebrating birthdays, anniversaries of self and significant others, vacations, bonding with family and friends, the achievements of self and significant others like getting good grades, a new job, or making some social contribution. Moreover, health-related issues like illness and accidents followed by family and relationship issues, and financial losses were also found prominently related to unhappiness in life. Based on the results of the qualitative analysis it's clear that there are distinct themes underlying the construct of HAW and happiness in personal life. People at work associate happiness with more professional reasons (e.g., salary, nature of work) as compared to the happiness in life where it is expressed mainly through social events and bonds. It gave us support to include only work-related aspects, reflected in 10-dimensions identified earlier, in our measure of happiness.

5 Study 2: Exploratory Factor Analysis (EFA)

5.1 Method

Based on the review of the literature and the results of study 1, the authors generated 65 items oversampling the identified ten dimensions (see Table 1). Both positive and negative items were generated for each dimension as per the criteria laid by Edwards (1983) and Hinkin (1995). The response option for items varied from 1 (strongly disagree) to 7 (strongly agree).

5.2 Sample

We collected data in two phases. In the first phase, the 65 item HAW scale was sent to 750 working people through hired field surveyors and volunteering undergraduate students of the University of Delhi. Total 314 subjects belonging to different sectors returned data of which 280 responses were found usable. In phase 2 the same questionnaire was personally administered to 270 working people out of which 259 responses were found usable. The total sample ($N = 539$) consisted of 46.2% working male and 53.2% females with the mean age and the mean work experience of 34.68 and 7.96 years respectively. The sample was majorly drawn from the Tier I Metro cities of India with 47.5% subjects belonging to Indian origin private sector organizations, 11.7% belonging to foreign origin private sector organizations, 17.1% belonging to public sector/government organizations, 1.7% entrepreneurs and remaining 1.1% per belonging to non-governmental organizations (NGOs).

5.3 Result

Before going for exploratory factor analysis (EFA), Velicer's minimum average partial (MAP) test was conducted to identify the reliable number of independent factors as it is a more objective method for identifying the possible number of factors as compared to its subjective counterparts like Kaiser's criterion or scree plot (O'connor 2000). Velicer's MAP test suggested four factors with eigenvalues ranging from 2.34 to 10.02. Furthermore, an initial reliability analysis with all 65 items confirmed sufficient reliability of the scale

($\alpha = .89$). Researchers have recommended considering the item-total correlation cut-off ranging from .3 (Nunnally and Bernstein 2010) to .4 (Edwards 1983; Field 2013) for the acceptance of items for further exploratory analysis. There were 42 such items retained after reliability analysis. There were no negative item-total correlations reflecting the absence of bad wording, sampling error or miskeying (Nunnally and Bernstein 2010). These 42 items were subjected to an iterative process of principal axis factoring which yielded four factors consisting of 16 items as shown in Table 2.

5.4 Results and Discussion

As per a priori conceptualization, contrasting dimensions emerged reflecting individual and organizational factors which jointly explained 55.62% of the variance. Based on the content of items organizational dimensions were named as supportive organizational experiences ($n = 4$ items, $\alpha = .73$) and unsupportive organizational experiences ($n = 3$ items, $\alpha = .72$), while employee dimensions were named as flow and intrinsic motivation ($n = 5$ items, $\alpha = .71$), and work repulsive feelings ($n = 4$ items, $\alpha = .75$). Furthermore, total score on the proposed scale was found significantly positively correlated with supportive organizational experiences ($r = .62$, $p < .01$) and flow and intrinsic motivation ($r = .65$, $p < .01$) dimensions, while being negatively correlated with work repulsive feelings ($r = -.68$, $p < .01$) and unsupportive organizational experiences ($r = -.47$, $p < .01$) dimensions. Again, correlation table (Table 3) shows that HAW correlates positively with the construct of positive emotion ($r = .39$, $p < .01$), satisfaction with life ($r = .24$, $p < .01$) and life orientation ($r = .15$, $p < .05$), and negatively with negative ffect ($r = -.24$, $p < .01$). These correlations are in conceptually intended direction and offer further supports to the validity of the proposed scale.

6 Study 3: Scale Validation Through Confirmatory Factor Analysis (CFA)

6.1 Method

Study 3 was conducted with an aim to confirm the factor structure of the proposed scale. Data was collected on the 16 item HAW scale along with four standardized validity measures whose details are given below in the measures section.

6.2 Sample

The sample consisted of 400 Indian working male (36%) and female (64%) with a mean age of 27.90 years and the average experience of 3.27 years. The respondents mainly belonged to private sector Indian origin organizations (68.2%), followed by public sector/government organization (20.5%), private sector multinational companies (11.4%). Around 45% respondents had a graduate degree, 35.6% reported having a master degree and rest reported diploma and other qualifications. Furthermore, 81.8% reported practising religion as Hinduism, 3.6% Islam, 3.6% Christianity, 3% Sikhism, 2.4% Jainism, and remaining 5.5% atheist and other religions.

Table 2 Results of EFA & CFA on retained items of the proposed scale

Items	Factor loadings based on principal axis factoring (Study 2, N = 539)				Standardized loadings (λ) based on confirmatory factor analysis of four-factor 12 item model (Study 3, N = 400)				Item reliability (λ^2)	Error term ($1-\lambda^2$)
	FIM	WRF	SOE	UOE	FIM	WRF	SOE	UOE		
i25. At my work, I remain inspired and try to inspire others as well (Gehrich 2012)[a].	**.690**	.075	.025	.030	.825				.681	.319
i27. I feel internally driven to do great things at my work (Hassanzadeh and Mahdinejad 2013; Zhang and Kemp 2009).	**.617**	−.038	.032	−.129	.856				.732	.267
i26. When I start doing my work I forget everything else (Kondo and Hirano 2016; Vanderkam 2015).	**.615**	−.094	−.093	.014	deleted				–	–
i24. I enjoy what I am doing at work (Revesencio 2015; Waterman 1993).	**.563**	−.003	.148	−.004	.765				.585	0414
i16. I continue to do a task till it is perfectly done (interpretation based on Bakker and Demerouti 2008).	**.491**	.094	.088	.094	deleted				–	–
i50. I am not very comfortable in approaching my boss.*	.004	**.635**	−.008	.034		.751			.564	.440
i57. I hate lot of people here for always being around the boss for personal gains (based on Chanania 2011; Schrage 2010).*	−.070	**.603**	.010	−.012		deleted			–	–
i32. I feel stressed at work (based on Kjerulf 2014; Nishitani and Sakakibara 2007).*	.053	**.564**	−.027	−.110		.855			.731	.269
i42. Often I feel like quitting my job (Coomber and Barriball 2007; Halbesleben and Wheeler 2008).*	.009	**.560**	−.020	−.065		.765			.585	.415
i31. My organization provides all necessary training and information to complete work on time.	.089	.118	**.643**	.086			.749		.561	.439
i41. The decision-making process in my company is fair and just (based on Martin et al. 1997).	−.056	−.083	**.632**	.012			.832		.692	.308

Table 2 continued

Items	Factor loadings based on principal axis factoring (Study 2, N = 539)				Standardized loadings (λ) based on confirmatory factor analysis of four-factor 12 item model (Study 3, N = 400)				Item reliability (λ^2)	Error term ($1 - \lambda^2$)
	FIM	WRF	SOE	UOE	FIM	WRF	SOE	UOE		
i21. Top leaders of my organization have clear vision and focus.	.051	−.053	**.628**	−.081			.713		.508	.492
i6. We celebrate and cheer each other at the accomplishment of targets.	.035	.021	**.583**	−.021			deleted		–	–
i19. My organization does not have proper guidelines to regulate team behaviour and work that require collective effort (based on Rook 1984).*	.054	.011	−.083	**.763**				**.756**	.572	.428
i11. My company does not have a proper interface that can allow us to work for social cause.*	−.076	.018	.097	**.665**				**.873**	.762	.238
i14. I don't get sufficient credit for my contributions.*	.023	.120	−.003	**.541**				**.717**	.514	.486
Eigen values	3.38	3.30	1.40	1.11						
% Variance explained (Total = 55.62%)	21.15	18.77	8.75	6.95						
α	.75	.71	.73	.72						

Items are presented with their original item number in parent scale. Deleted—item deleted during confirmatory analysis to improve the model fit

FIM flow and intrinsic motivation, *WRF* work repulsive feelings, *SOE* supportive organizational experiences, *UOE* unsupportive organizational experiences

Primary factor loadings are shown in bold

* Denote the negatively worded items of the scale

[a] Some studies supporting a reflective influence of happiness at work on the content measured in items are shown in parentheses. The detailed explanation of the same can be seen in the discussion part

Table 3 Correlation among psychological variables (Study 3)

			1	2	3	4	5	6
1	Happiness at work scale	r	$\alpha = .72$					
		N	398					
2	Positive affect	r	.394**	$\alpha = .83$				
		N	173	174				
3	Negative affect	r	−.243**	−.343**	$\alpha = .74$			
		N	169	170	170			
4	Positive life orientation	r	.151*	.314**	−.310**	$\alpha = .43$		
		N	174	174	170	175		
5	Satisfaction with life	r	.239*	.214*	−.326**	.081	$\alpha = .75$	
		N	108	108	105	109	109	
6	Oxford happiness scale	r	.527**	–	–	–	–	$\alpha = .79$
		N	137	–	–	–	–	137

Values on diagonal denote Cronbach's Alpha (α) value of the respective scales. Oxford Happiness Scale was not administered in Study 3 hence its correlation with other scales are not available

* $p < .05$; ** $p < .01$; r denotes correlation and N denotes respective sample size

6.3 Measures

6.3.1 *Oxford Happiness Questionnaire*

Oxford Happiness Questionnaire (Hills and Argyle 2002) was used to measure the uni-dimensional subjective well-being factor among subjects. It is a 29 item Likert- type scale ($\alpha = .79$) with response options varying from 1("Strongly disagree") to 6 ("Strongly agree"). The scale contains 12 negative items all of which were reverse scored before analysis. A sample item of the scale is "I feel that life is very rewarding."

6.3.2 *Positive and Negative Affects Schedule (PANAS)*

To measure the affective state of the subjects we used 20-items PANAS (Watson et al. 1988) that contains 10 adjectives measuring the positive affect (e.g., inspired, enthusiastic; $N = 175$, $\alpha = .83$) and 10 adjectives measuring negative affect (e.g., irritable, distressed; $N = 174$, $\alpha = .74$). Subjects were asked to indicate the extent of feelings on these adjectives on a 5-point Likert-type scale varying from 1 ("Not at all") to 5 ("Extremely").

6.3.3 *Satisfaction with Life Scale (SWLS)*

It is five-item Likert-type positively worded scale to measure global life satisfaction (Pavot and Diener 1993) with response options varying from 1 ("Strongly disagree") to 7 ("Strongly agree"). A sample item of this scale ($N = 109$, $\alpha = .75$) is "I am satisfied with my life."

6.3.4 Revised Life Orientation Test (LOT-R)

This scale measures dispositional optimism defined in term of generalized outcome expectations (Scheier et al. 1994) based on 10 Likert-type items scored from 0 ("Strongly disagree") to 4 ("Strongly agree"). The scale included three filler items which were not included in the analysis. Although its reliability was found comparatively low in current sample ($N = 175$, $\alpha = .43$) but Gustems-Carnicer et al. (2016) have reported overall good reliability ($N = 291$, $\alpha = .72$) for the Spanish version of this scale.

6.4 Results and Discussion

6.4.1 CFA

We performed CFA using AMOS 23 to validate the factor structure that emerged after EFA in the study 2. For testing the model fit we relied on the values of the recommended indices of model fit, i.e. relative Chi square (χ^2/df) value between 1 and 3, Goodness of Fit Index (*GFI*) and Comparative Fit Index (*CFI*) value of more than .95, Standardized Root Mean Square Residual (*SRMR*) values less than .08, and Root Mean Square Error of Approximation (*RMSEA*) value less than .06 (Hu and Bentler 1999). Since the models were non-nested, we used Akaike Information Criterion (*AIC*) to compare the relative quality of competing models. *AIC* is probably the best and most widely known and used (Cavanaugh 2012) parsimony-based criterion that indicates loss of information while selecting a candidate model as the true model (the actual model that give rise to sample data). A model having a lower value of *AIC* is preferable as it indicates less information loss as compared to the true model. We calculated *AIC* change (Δi) defined as the difference between AIC value of a competing model (*ith* model) and the minimum AIC value reported for best fitting model (i.e., $\Delta i = AIC_i - AIC_{min}$) and ranked models based on their AIC value in the ascending order (model with lowest AIC change value has been given the rank of 1).

To allow free estimation of factor loadings for all the four EFA dimensions and keep the model identifiable the variance of second order HAW construct was constrained to 1 (Model 6.3). The initial analysis (Model 6.3) showed mediocre model fit ($\chi^2/df = 3.335$, $GFI = .904$, $CFI = .921$, $SRMR = .074$, $RMSEA = .077$, $AIC = 405.539$). To improve the model fit we analyzed the correlation between error terms and based on the modification indices and low standardized loadings we iteratively eliminated four items (i6, i16, i26, i57) to avoid redundant items and improve model fit. This offered 12-item hierarchical model (Model 6.2) with acceptable fit ($\chi^2/df = 1.923$, $GFI = .962$, $CFI = .978$, $SRMR = .054$, $RMSEA = .048$, $AIC = 152.154$). We further compared this model (Model 6.2) with a 16-item (Model 7.1) and 12-item (Model 7.2) reflective four-factor model in which the second-order happiness construct, whose variance was constrained to 1, was eliminated because a model with no imposed constraints with all factor loadings varying freely is preferable as compared to a constrained model (Dahling et al. 2008). The 12-item four factor reflective model (Model 7.2) showed better fit as compared to the 12 item reflective hierarchical (Model 6.2) model ($\chi^2/df = 1.635$, $GFI = .969$, $CFI = .985$, $SRMR = .039$, $RMSEA = .040$, $AIC = 135.55$). We found the most robust indices for the Model 7.2, thus, compared it with other competing models. The competing models included a null model, a common factor model, three pairs of reflective versus formative models with reflective PANAS (Model 3.1 and Model 3.2), reflective SWLS (Model 4.1 and Model 4.2), reflective LOT-R (Model 5.1 and Model 5.2), a formative first-order hierarchical model (Model 6.1), and a 16-item four factor reflective model (Model 7.1). The results of CFA along with indices of model fit are presented in Table 4.

Based on the results of comparative model fit we accepted 12-item four-factor reflective Model 7.2 (Fig. 2) as it shows better indices of model fit as compared to all other models. Furthermore, all reflective models performed better as compared to their formative counterpart.

6.4.2 Reliability, Convergent and Discriminant Validity

Table 5 shows that the proposed scale has no convergent and discriminant validity issues. All the composite reliabilities and coefficient *H* values are greater than .8 reflecting the good reliability of the constructs. All the AVE values are more than .5 reflecting convergent validity. Furthermore, MSV values are less than their AVE values of the respective constructs along with square root of AVE values being more than off-diagonal values indicating the discriminant validity of the constructs.

7 General Discussion

Our study shows that people do distinguish between happiness in personal and professional life (study 1). We further identified four major dimensions of the happiness at work (study 2) and used a series of 13 competing models to confirm the four-factor structure (Study 3) of the proposed scale. Scale development in psychology essentially involves working with reflective constructs where constructs reflect the indicators in contrast to index development exercise, which is more common in disciplines like economics and sociology, with formative constructs (Diamantopoulos and Siguaw 2006; Diamantopoulos and Winklhofer 2001). Thus, measurement of psychological constructs like happiness, attitude, personality is done in a way where indicators (manifestations) typically originate from their parent construct (latent variable). The four EFA dimensions are essentially latent factors where items have been conceptualized to originate from their respective factors. However, on the surface level, some indicators give the impression of being a formative measure of happiness at work. Technically the major points of distinction between formative and reflective models are that reflective models have indicators which are interchangeable, correlated, thus reflect internal consistency, along with having error terms and forming statistically identifiable models (Diamantopoulos and Winklhofer 2001, p. 271). The obtained exploratory model meets the interchangeability criterion as removal of any item will not lead to any substantial change in the nature of underlying construct (although it may affect construct reliability or may create the problem of model identification). Similarly, there is an acceptable amount of correlation between the four dimensions and internal consistency shown by the final scale.

However, the relationship between variables is not always possible to clearly categorize as either formative or reflective because casual relations among psychological variables can sometimes work mutually. For example, volunteering (time) and donation (money) have been found to bring greater well-being, and people with a greater sense of well-being have been found to spend more hours in volunteering and donate more money (Aaker and Akutsu 2009; Thoits and Hewitt 2001). However, we accept the premise that psychological measures are essentially reflective and, keeping in mind certain limitations of this dichotomy, we discuss the obtained results about the content and factor structure of the

Table 4 Fit indices for test of discriminant validity (N = 400) (Study 3)

Model	Second order	First order	χ^2	*df*	χ^2/df	SRMR	RMSEA	GFI	CFI	*AIC*	$\Delta i = AIC_i - AIC_{min}$	Model rank
Model 1: Null model	–	–	2136.914***	66	32.377	–	.280	.431	.000	2160.914	2025.364	13
Model 2: Common factor model	–	Reflective	1099.863***	54	20.368	.146	.220	.638	.495	1147.863	1012.313	10
Model 3.1	PANAS reflective	Four factor formative	1248.689***	464	2.691	.094	.125	.693	.906	1612.935	1477.385	12
Model 3.2	PANAS reflective	Four factor reflective	781.602***	462	1.692	.089	.080	.708	.962	1157.230	1021.680	11
Model 4.1	SWLS reflective	Four factor formative	229.089***	119	1.925	.177	.093	.795	.781	297.089	161.539	6
Model 4.2	SWLS reflective	Four factor reflective	207.912***	115	1.808	.1403	.086	.822	.815	283.912	148.362	5
Model 5.1	LOT-R reflective	Four factors formative	298.025***	90	3.311	.212	.115	.808	.740	358.025	222.475	8
Model 5.2	LOT-R reflective	Four factors reflective	159.430***	86	1.854	.092	.070	.894	.908	227.430	91.880	3
Model 6.1	HAW reflective	Four factor formative	287.362***	54	5.322	.206	.104	.882	.887	335.362	199.812	7
Model 6.2: Hierarchical model	HAW reflective 12-item	Four factor reflective	96.154***	50	1.923	.054	.048	.962	.978	152.154	16.604	2
Model 6.3	HAW reflective 16-item	Four factor reflective	333.539***	100	3.335	.074	.077	.904	.921	405.539	269.989	9
Model 7.1	–	Four factor reflective 16-items	182.990***	94	1.947	.067	.049	.948	.970	266.990	131.440	4

Table 4 continued

Model	Second order	First order	χ^2	*df*	χ^2/df	SRMR	RMSEA	GFI	CFI	*AIC*	$\Delta i = AIC_i - AIC_{min}$	Model rank
Model 7.2: Accepted model	–	Four factors reflective 12-items	78.483**	48	1.635	.039	.040	.969	.985	135.55 (AIC_{min})	0	1

** $p < .01$; *** $p < .001$

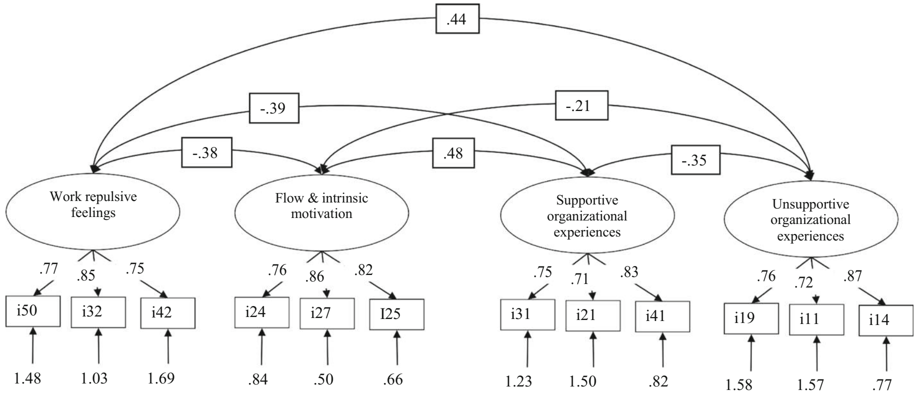

Fig. 2 Results of Confirmatory Factor Analysis of the Happiness at Work Scale

Table 5 Reliability, convergent and discriminant validity

	Scale dimensions	CR	AVE	MSV	Coefficient *H*	1	2	3	4
1	Flow and intrinsic motivation	0.856	0.666	0.232	0.862	0.816			
2	Supportive organizational experiences	0.809	0.587	0.232	0.820	0.482***	0.766		
3	Unsupportive organizational experiences	0.827	0.616	0.196	0.848	−0.207***	−0.348***	0.785	
4	Work repulsive feelings	0.834	0.627	0.196	0.844	−0.377***	−0.388***	0.442***	0.792

CR composite reliability, *AVE* average variance extracted, *MSV* maximum shared variance

*** $p < .001$

proposed scale in the following paragraphs. Although we have taken special care to incorporate only researches linking the content of items with workplace happiness as far as possible, due to the paucity of the researches in this direction researches related to general happiness have also been incorporated wherever relevant.

7.1 Flow and Intrinsic Motivation

Researchers have studied the relationship between happiness and inspiration (*item 25*) in both formative and reflective manner. For example, Thrash et al. (2010) found that inspiration significantly enhances well-being, while Chaiprasit and Santidhiraku (2011) could not find a significant influence of job inspiration on HAW ($B = .13$, $t = .16$, $p = .11$). Gehrich (2012) suggest a reflective influence of happiness on inspiration and holds the view that “everytime employees experience a small burst of happiness, they get primed for creativity and innovation. Happy people are more inspired and productive”

(p. 98). Furthermore, researchers have reported a significant relationship between happiness and achievement motivation (Hassanzadeh and Mahdinejad 2013), and intrinsic motivation ($r = .31$, $p < .05$, Zhang and Kemp 2009). Thus, a high level of HAW will be reflected in internal drive and motivation of the employees to do great things at work (*item 27*).

HAW is also reflected in the extent to which employees feel absorbed in their work (*item 26*). Early researchers emphasized a formative relationship between feeling absorbed in work and happiness, for e.g., Myers and Diener (1995) believe that either at work or leisure people enjoy themselves more when they are absorbed in the flow of an activity. However, a recent article in Fortune (Vanderkam 2015) while citing Japanese authors (Kondo and Hirano 2016) says that happy people actively seek work that can absorb them. It doesn't appear logical to presume that a passive absorption (which may happen during the meeting of deadlines or piling up of work) will always lead to HAW. Furthermore, recent researches have shown that when employees are happy they tend to enjoy work more (*item 24*) and are more effective, creative and collaborative (Revesencio 2015). Furthermore, perfectionist people are rarely happy as things rarely reach to the state of perfection or stay there (Matthews and Ewens 1988; Walsh 2013). This diminishes the scope of the formative effect of perfectionist attitude on happiness at work. If employees are happy, they reflect greater engagement with work and organization due to the influence of positive mood which is associated with happiness at work. As happy people show increased sensitivity to opportunities at work, are more helpful, confident and optimistic (Bakker and Demerouti 2008) they are more likely to strive for perfection in their work (*item 16*).

7.2 Work Repulsive Feelings

The simple idea of 'approaching boss' for anything is enough to cause discomfort and anxiety among most of the employees unless employees prefer socializing with the boss, developing personal equations to receive guidance or improve their power and position in the organization. Meeting with the boss unavoidably involves indulging in some normative or expected behaviour, or one that pleases the boss along with the possibility of being judged accordingly as any type of organization (structure) almost always gives more power to the bosses than the employees. This is likely to create anxiety and unhappiness among the employees. According to a Harvard Business review article (de Vries 2016) around half of the American employees leave their jobs at some point in their career to get away from their bosses. Similar or even higher figures can be expected in other continents of the world. Thus, unhappiness at work could be well reflected in the form of the inhibition to approach or tendency to avoid one's boss (*item 50*). Being uncomfortable in approaching one's boss or tendency to avoid the boss as far as possible could be a good behavioural measure as well as a heuristic to understand unhappiness at work. However, here other variables may also play a role, for e.g., personality traits like introversion, psychological barrier, perceived negative attitude or behaviour of the boss, which is not uncommon, like micromanaging, bullying, stealing credit, poor listening habit, and negative past experiences. Furthermore, if employees (either extrovert or introvert) have no choice in choosing the way they become happy, then the scenario is likely to bring only unhappiness (Hills and Argyle 2001). As employees rarely have choices or equal power when they approach their bosses hence a discomfort to approach one's boss can be considered a good indicator of employee unhappiness.

Furthermore, it is believed that introverts comprise over one-third of the population but the actual number could be between 47 and 55% (Helgoe 2013). According to another estimate the prevalence of introversion and extroversion among the bosses is in the ratio of 50:50 (Brush 2012) so there is always more than the chance possibility of unhappiness at work being reflected by a discomfort in approaching one's boss who could be a person of opposite personality disposition. Although extroverts consistently rate themselves higher on happiness but that may be due to their self-serving bias and perception of being fit in a seemingly extroversion-endorsing culture (Hills and Argyle 2001).

Bosses may knowingly or unknowingly promote dysfunctional behaviour at work like sycophancy, suck-ups, the motivated admiration which may generate repulsive or loathsome feeling among other employees (Schrage 2010). In the presence of such dysfunctional behaviours in which honest and task oriented employees feel incapable of indulging, the unhappiness of employees would be reflected through the feeling of nausea and repulsiveness toward sycophant employees in particular (*item 57*) and work in general. The intensity of unhappiness brought by this dilemma, especially when professional losses could be up to 5% in this competitive world of work (Chanania 2011), would be reflected in the feeling of disgust which is a strong primary visceral emotion. Since prevalence rate of sycophancy is very high in the organizations, up to 80% in case of public sector organizations (Chahal and Poonam 2015), unhappiness at work would often be reflected through the feeling of repulsiveness in such situations.

Though stress may appear to have a formative influence on happiness, it is more prudent to think that absence of HAW will be reflected through symptoms like stress (*item 32*), sleep disturbances, poor health, burnout, depression, poor productivity or similar negative outcomes. Around 20 to 40% employees feel unhappy and miserable at work (Kjerulf 2014) which has been found to be positively correlated with unhealthy eating behaviour, stress, and depression (Nishitani and Sakakibara 2007). The experience of unhappiness at work seriously affects the attitude of employees toward their job which could be reflected in employee's intention of quitting the job (*item 42*) but this may not result in the employee actually quitting the job as the relationship between attitude and behaviour is essentially weak (Erwin 2014). This also explains why a large number of people don't quit their job even after being unhappy and unengaged to it. Feeling of unhappiness at work brings disengagement and lack of embeddedness that leads to intention to quit (Halbesleben and Wheeler 2008). Unhappiness with work also means dissatisfaction at work that again leads to intention to quit (Coomber and Barriball 2007; Scott et al. 2003).

7.3 Supportive/Unsupportive Organizational Experiences

Items 31, 41, 21 (belonging to supportive organizational experiences dimension) and items 19, 11, 14 (belonging to unsupportive organizational experiences dimension) essentially reflect how happiness affects the employee evaluation of various types of organizational support or lack of it. The general premise underlying these items is that happiness at work will lead to positive evaluation of various organizational supports while unhappiness will lead to negative evaluation reflected in perceived lack of support from the organization (Batra and Stayman 1990; Martin et al. 1997). Happy employees are more likely to express satisfaction with the company training programs (*item 31*) and the organizational support that is required to complete their work. Happy employees are more likely to value the learning of new skills and will show greater keenness to undergo training programs so as to improve their skills, self-esteem, and earnings. It is not uncommon to observe happy and successful athletes offering a more favourable evaluation of their coaches or happy students showing better class attendance, class participation, appreciation for the teacher, and overall satisfaction with the teaching–learning process.

Moreover, the perception of fairness at work again to a large extent depends on how employees feel at work (*item 41*). A happy and satisfied employee is more likely to perceive decision-making process as fair and just because people in positive mood offer more favourable evaluation than people in negative mood (Martin et al. 1997). Similarly, happy employees are more likely to hold a favourable evaluation about their organization's vision and focus (*item 21*). Furthermore, a happy employee is less likely to be jealous of achievement of other colleagues, and more likely to participate in celebrating and cheering others/teams at the accomplishment of targets (*item 6*). We consider it one of the important indicators of employee happiness expressed at group or team level. Celebrating and cheering others at the accomplishment of targets is an indication of the extent of mood-sharing in the organizations which is positively related to shared vision and organizational engagement (Mahon et al. 2014).

In contrast to this, unhappiness at work will be reflected in various sorts of unfavourable evaluation by the employees for organizational team behaviour (*item 19*), avenues for doing socially impacting work (*item 11*) and stealing of work credit by others (*item 14*). It's not uncommon to find unhappy employees being less participative in teams, complaining about politics and stealing of credit by others, and looking for alternative avenues to do meaningful work. Modern workplaces strongly favour team culture and sociability offering enough scope for developing professional and personal relations which may also bring workplace politics, favouritism, and a host of counterproductive work behaviours (Morrison and Nolan 2007). Since negative social outcomes are more strongly related to individual well-being than positive social outcomes (Rook 1984) it will lead affected employees to hold a negative attitude or, if possible, avoid working in teams especially when they perceive team behaviour is not properly regulated (*item 19*), or others might steal the credit of their work (*item 14*), or the work is not meaningful or socially impacting (*item 11*).

In addition to this, results of the confirmatory analysis (Table 4) showed good model fit for the four-factor reflective model (refer Model 7.2 in Table 4) as compared to all other competing models. Hence, based on the statistical criteria for reflective models prescribed by Diamantopolous et al. the items retained after EFA are essentially parts of the reflective constructs of happiness at work.

8 Application, Limitations, and Future Directions

8.1 Application

The four dimensions could be arranged in the form of a happiness quadrant giving four distinct happiness styles (as shown in Fig. 3) based on the interaction between the quantum of positive and negative characteristics of individual and organizations on a two-dimensional continuum. The empirical validation of these happiness styles will require a separate research which we would like to take up in future.

A combination of the supportive work experiences and flow and intrinsic motivation can bring flourishing style of happiness in which both individuals and organizations grow and prosper. However, when an intrinsically motivated employee has to work in an unsupportive workplace s/he accumulates frustration and unhappiness leading to a frustrating

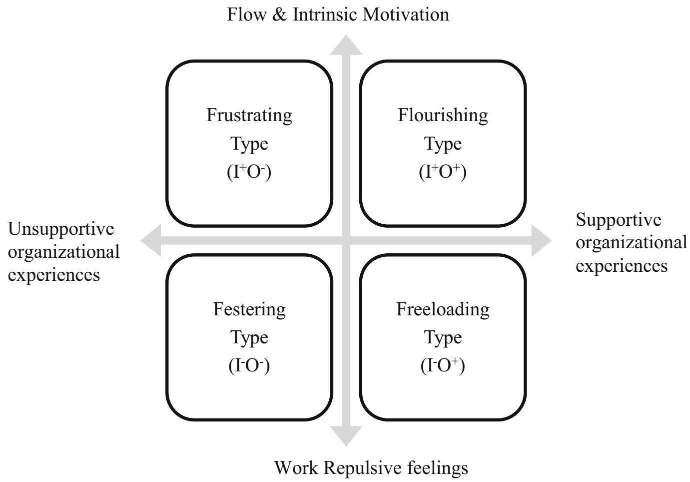

Fig. 3 Four predicted styles of happiness at work. *I* Individual characteristics, *O* organizational characteristics. X-axis represents organizational continuum while Y-axis represents individual continuum. *Plus sign* (+) shows positive characteristics while *minus sign* (−) shows negative characteristics

style. Contrary to this, when organizational experiences are supportive but employees dislike their job then it leads to a freeloading, parasitical type of situation in which organizations will be the main loser. Lastly when organizational experiences are unsupportive, and employees also dislike their jobs then we get a mutually degenerating, festering style in which neither employees nor individuals benefit.

8.2 Limitations and Future Directions

The broad deductive framework of HAW offered by current research is simple and appealing. In future researches, the scale and the proposed happiness styles need to be validated by undertaking cross-cultural mixed method studies. A common limitation of psychometric studies is the use of questionnaire data which can also be a limiting factor for the current study. In future experiments and interventions can be planned on the identified dimensions to offer experimental validation for the scale.

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